

Screens — Helix Console (admin web/desktop app)

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Master technical specification — Volume 10 (Design System), nano-detail deep-dive. This document **owns** the complete screen inventory and per-screen UX specification of **Helix Console** — the HelixVPN administration app (Flutter, flavor: HelixFlavor.console, capability {Capability.admin}, **no helix_core_ffi**, the **only** flavor that builds to Web) [03-client-core-and-ui.md §6, CI3]. It defines, screen by screen: purpose, desktop-first responsive layout, the helix_design components used, every state (loading / empty / error / partial / live), navigation, interactions, accessibility, and the **light + dark** rendering — each backed by an ASCII wireframe.

SPEC-ONLY. It describes *what each screen is and how it behaves* — not the shipping console build. Original HelixVPN UX design work; the control-plane data each screen surfaces is owned by the cited control-plane docs.

Boundary with sibling docs. Owns: the Console screen set + per-screen layout/state/interaction/all. **Consumes:** the colour roles + connection / feedback semantics + per-app **violet** accent [color-system.md §2/§6]; the token tiers + spacing/radius/breakpoint scales [design-tokens.md §6]; the signature components (AdaptiveScaffold, NetworkTile, PolicyEditor, ...) [component-library.md, sibling, this wave]; the OpenDesign engine that emits the tokens [opendesign-foundation.md]; the **Connector** counterpart screens [screens-connector.md, sibling]; the **Client** screens [screens-client.md, sibling]. The **data** every screen renders is owned by the control-plane: [02-control-plane.md] + the V3 service docs

[v03-control-plane/svc-registry.md], [.../svc-policy.md], [.../svc-coordinator.md], [.../svc-telemetry.md], [.../svc-ipam.md], [.../svc-identity.md].

Evidence base. [CP §N] = final/02-control-plane.md; [CLIENT §N] = final/03-client-core-and-ui.md; [REG §N]/[POL §N]/[COORD §N]/[TEL §N]/ [IPAM §N] = the matching final/v03-control-plane/svc-*.md; [COLOR §N] = final/v10-design/color-system.md; [DT §N] = final/v10-design/design-tokens.md. Claims not grounded in the evidence base or in this document's own original design choices are tagged UNVERIFIED per constitution §11.4.6 — never fabricated. Layout/state/interaction designs are **original HelixVPN UX work**.

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0. What the Console is — and the design principles that shape every screen

Helix Console is the **admin brain's front end**: it is where a tenant administrator manages *who may reach what*. It is the UI over the Go control plane's REST + WebSocket/SSE surface [CP §8] — it **never** opens a tunnel, never links the Rust core

(Capability.admin, no helix_core_ffi, CI3 [CLIENT §6]) — so it is the one flavor that builds to **Web** and also ships as a desktop app (macOS / Windows / Linux from the same tree).

Five principles govern every Console screen:

1. **Density without clutter — desktop-first.** The Console is a power tool used on lg (1024) and xl (1440) viewports [DT §6.6]; it presents heavy tables, a graph, and a code-like policy editor. It uses the **expanded AdaptiveScaffold** (extended NavigationRail + multi-pane) [CLIENT §7.3] and degrades gracefully to a phone layout, but the home turf is the wide screen.
2. **Authority accent = violet, safety colours unchanged.** The Console's brand-overridable accent is violet.* [COLOR §6] — it reads as an admin/authority surface, distinct from the Client's indigo and the Connector's teal. The **connection-state and feedback semantics never change** (D-COLOR-1): a device shown "online/protected" is the same green everywhere.
3. **Live, not polled.** Device lists, the topology graph, and the audit feed fold GET /v1/stream WS/SSE events (device.online, route.changed, policy.compiled, device.revoked) into state and update **without a manual refresh** [CLIENT §8.3, CP §8]. A small live-connection indicator tells the admin the stream is healthy.
4. **Fail-closed, preview-before-apply.** The highest-stakes screen (Policy editor) compiles **dry-run first** and shows an effect-diff before anything activates [POL §4/§5]; destructive control actions (revoke, deactivate) require explicit confirmation per §11.4.66.
5. **Privacy by construction is visible.** The Console **cannot** show traffic / connection / flow logs because they do not exist (C3 [CP §0.1]). The Audit screen states this plainly — it shows *control actions*, never user traffic.

Honest boundary (§11.4.6). This document specifies layout, state, and interaction. It does **not** re-specify the REST/WS payloads (owned by [CP §8] + the V3 service docs) nor the token hex/contrast (owned by [COLOR]/[DT]). Where a concrete control-plane field name is cited it is grounded in those docs; a value not so grounded is marked UNVERIFIED.

1. The Console shell (AdaptiveScaffold, responsive law, command palette)

Every screen renders **inside** the Console shell. The shell is AdaptiveScaffold [CLIENT §7.2] branching on **width, never Platform.isX** [CLIENT §7.3]:

Breakpoint [DT §6.6]	Width	Shell layout
compact	< 600	BottomNavigationBar (5 primary destinations) + single pane; overflow destinations in a "More" sheet
medium	600–1024	NavigationRail (icons + labels) + master/detail two-pane
expanded	> 1024	Extended NavigationRail (icon + text, pinned) + multi-pane (list detail live-events rail) — the Console home turf

Helix Console	[acme-corp ▼]	%K	● theme	●
live	○ admin	← top app bar (surface.default, violet brand mark)		
■ Dashboard				LIVE
EVENTS				
■ Devices		(primary content pane)		
○ Users & SSO				12:04
dev on				
□ Policy				12:03
policy				
◆ Topology				
v8 act.				
■ IPAM				12:01
dev off				
≡ Audit				
...				
▲ Telemetry				
(folds /v1/				
■ Billing				
stream WS)				
⚙ Settings				
● stream OK				

extended rail content (master|detail)
live-events rail

- **Top app bar** carries: the Console brand mark (violet), the **tenant switcher** [acme-corp ▼] (an admin with rights to >1 tenant; one tenant ⇒ static label), the **%K command palette** trigger, a **theme toggle** (● system/light/dark), the **live-stream indicator** (● green = stream healthy / amber =

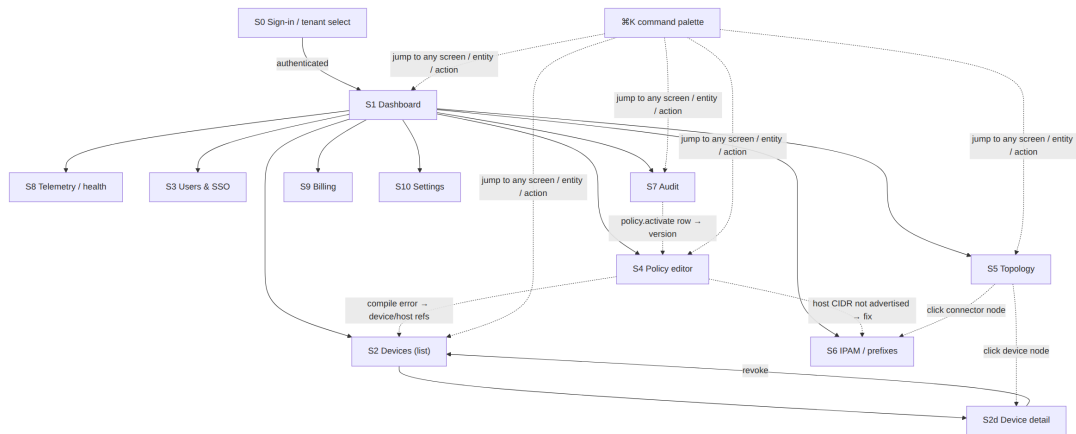
reconnecting / red = stream down — reuses the connection-state palette [COLOR §3]), and the signed-in admin avatar + role chip.

- **Command palette (§K / Ctrl-K)** [CLIENT §7.3] — fuzzy jump to any screen, device, user, policy version, or action ("revoke device ...", "activate policy v7"). Keyboard-first; every action it lists is also reachable by mouse.
- **Live-events rail** (expanded only) folds the WS/SSE stream; on medium/ compact it collapses into a bell/notification sheet.
- The **stream-health chip** at the rail foot is the §11.4.6-honest signal that "live" is real: amber when the WS reconnects, red + a banner when it is down (the lists then show a "may be stale — reconnecting" notice rather than lying).

2. Screen inventory + navigation map

#	Screen	Route	Primary data source	Min role [CP §8.1]	Key components
S0	Sign-in / tenant select	/signin	identity (OIDC) [svc-identity.md]	(unauth)	brand splash, OIDC button, tenant chooser
S1	Dashboard / overview	/	aggregate of registry + telemetry + policy	member+	stat cards, SLO sparkline, live-event digest
S2	Devices — list	/devices	GET /v1/devices [REG §3]	member+	data table, presence dot, filter bar
S2d	Device — detail	/devices/:id	device + certs + presence [REG §2/ §5]	member+	detail header, cert panel, revoke action
S3	Users & SSO	/users	identity / OIDC config [svc-identity.md]	admin	roster table, role editor, OIDC config form
S4	Policy editor	/policy	policies.spec + compiler [POL §2/ §4]	operator+	PolicyEditor, effect-diff, version history
S5	Topology	/topology	coordinator graph [COORD §1]	member+	graph canvas,

#	Screen	Route	Primary data source	Min role [CP §8.1]	Key components
S6	IPAM / prefixes	/ipam	overlay pool + advertised_prefixes [IPAM §1, REG §6]	operator+	node/edge inspector pool card, prefix table, conflict list
S7	Audit	/audit	audit_events [TEL §4]	admin (read)	audit feed, action filter, no-traffic banner
S8	Telemetry / health	/health	Prometheus series [TEL §3]	operator+	SLO gauges, convergence histogram, alert list
S9	Billing / subscription	/billing	billing provider UNVERIFIED	admin	plan card, seat usage, invoices
S10	Settings	/settings	tenant config + per-user prefs	varies	tenant settings, theme, session, danger zone



The map is a hub-and-spoke from the Dashboard, with **cross-links that follow the data**: a policy compile error that names an un-advertised host CIDR links to IPAM (S6); a topology connector node links to its IPAM prefixes; an audit policy.activate row links to that policy version in the editor.

3. Sign-in (OIDC + tenant select)

Purpose. Authenticate a tenant admin (OIDC session) and, when the identity maps to more than one tenant, choose which to administer. Managed (OIDC) is the Console's identity mode [CP §9.1] — the anonymous device-token path is a *Client* enrollment concern, not a Console sign-in.

Layout (desktop-first, centred card).

Console
mark
network

◆ Helix

Administer your private

← violet brand

Continue with your identity provider

⦿ Sign in with SSO (OIDC)

Self-hosted? Enter your control-plane URL

https://cp.acme.example

action.primary (violet)

input

No traffic is ever logged. Helix stores identity & policy only.

← privacy note (C3)

Tenant select (post-auth, multi-tenant identity).

Choose a tenant

◆ acme-corp	142 dev	← selectable rows; presence
◆ acme-eu	38 dev	
◆ lab	6 dev	

States. *Loading* — brand splash + indeterminate progress while the OIDC redirect resolves. *Empty* — identity maps to **zero** tenants ⇒ a clear "your account is not an admin on any tenant; contact your operator" message (not a spinner that hangs). *Error* — OIDC failure shows the honest provider reason (feedback.error, [COLOR §2.4]) with a retry, never a generic "something went wrong"; an unreachable control-plane URL is a distinct, actionable error.

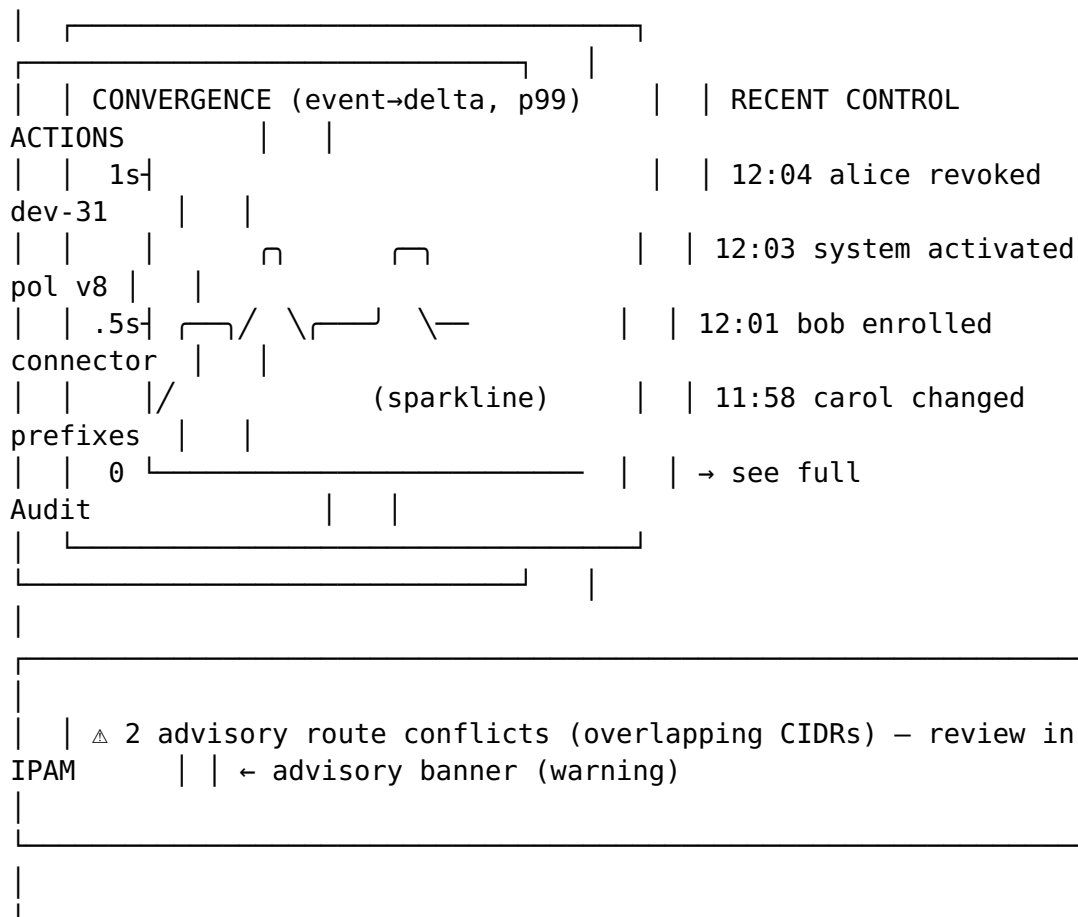
Interactions / a11y. SSO button is the default focus + Enter-activates; the control-plane-URL field validates as a URL on blur. Full keyboard nav; the brand mark has an alt. **Light/dark:** the card is surface.raised, the SSO button action.primary (violet #6D28D9 light / #A78BFA dark, both AA-proven [COLOR §6]); the privacy note is text.secondary.

4. Dashboard / overview

Purpose. The at-a-glance health of the tenant's network: how many devices are online, whether the active policy is healthy, whether the control plane is meeting its < 1 s convergence SLO [CP §10.2], and a digest of recent control activity. It is a **read** screen (member+) that links into every deeper screen.

Layout (expanded, three-zone).

Dashboard – acme-corp				updated
live ●				
DEVICES	CONNECTORS	ACTIVE	CONVERGE	←
stat cards				
142	7	POLICY	p99	
(elevation.card)				
118 online	6 online	v8 ✓	0.42 s ✓	
▲ green	▲ green	green	green	



Components. Four **stat cards** (elevation.semantic.card [DT §6.3]) each with a value, a sub-count, and a state-coloured indicator dot; a **convergence sparkline** sourced from `helix_reconcile_seconds p99` [TEL §3.2, COORD §7.3]; a **recent-actions digest** (last `N audit_events` [TEL §4]); an **advisory banner** summarising open `route.conflict.detected` items [REG §6.3] (advisory, never blocking — links to IPAM S6).

States. *Loading* — skeleton cards (shimmer placeholders, never a blank screen). *Empty* — a freshly-bootstrapped tenant with 0 devices shows an onboarding card: "Enroll your first connector" → mint an enroll token (links to Devices/enroll). *Error* — if an aggregate query fails, that one card shows an inline error + retry while the rest render (no all-or-nothing). *Live* — counts and digest update from `/v1/stream` without refresh; the "updated live ●" chip goes amber if the stream drops.

SLO colour logic (honest, §11.4.6). The convergence card is green only when `p99 < the SLO target (1 s [CP §10.2])`; amber within 1-2×; red above — and the card states the *measured* number, never "healthy" without the figure.

A11y / light-dark. Stat values are text.primary (AAA both themes); the indicator dots carry a text/icon label too (never colour-only, CI5 [CLIENT §0.1]). Cards are surface.raised; the warning banner is feedback.warning (amber.700 light / amber.400 dark [COLOR §2.4]).

5. Devices (list · detail · approve / revoke)

5.1 Devices — list (S2)

Purpose. The roster of every device (clients **and** connectors — one devices table, kind discriminates [CP §2.2, REG §2]). Filter, sort, search, and drill in. Live presence from Redis TTL keys via the stream [REG §5.3, TEL §5].

Layout (data table + filter bar).

Devices (142) [search name / overlay-ip] [+ Enroll device ▼]					
Filter: ● all ○ clients ○ connectors Status: ○ online ○ offline ○ all					
●	NAME	KIND	OVERLAY IP	OS	
LAST SEEN					
●	alice-mbp	client	fd7a:helix:1::2	macos	
online	← green presence dot				
●	warehouse-gw	connect.	fd7a:helix:1::5	linux	
online					
○	bob-pixel	client	fd7a:helix:1::9	android	3 h ago
	← grey (offline)				
○	old-laptop	client	fd7a:helix:1::4	windows	
revoked	← red (revoked)				
...					
118 online · 23 offline · 1 revoked < 1 2 3 ... >					
rows: 25 ▼					

Interactions. Click a row → device detail (S2d). The [+ Enroll device ▼] split-button mints a single-use enroll token [CP §9.2] and shows it as text + QR (operator hands it to the device); the QR/token is **never** logged (§11.4.10). Column sort; multi-facet filter;

debounced search over name + overlay IP. Presence dot uses the connection-state palette: green online / grey offline / red revoked — **always paired with the text status** (CI5).

States. *Loading* — table skeleton rows. *Empty* — "No devices yet — enroll your first connector or client" with the enroll CTA. *Error* — a banner ("couldn't load devices — retry") above the last-known rows (stale, marked). *Live* — device.online/.offline/.revoked/.enrolled events mutate rows in place; a newly enrolled device animates in (motion.stateXfade [DT §6.4]).

5.2 Device — detail (S2d)

Purpose. Everything known about one device (identity only, never traffic): identity, overlay IP, presence, current transport/RTT (presence/health only, C3), certificate state, group memberships, and the **revoke** action. For a connector it also surfaces its advertised prefixes (links to IPAM).

< Devices		alice-mbp	● online	
[Revoke ∅]				
Identity				
Certificate				
Kind	client	Serial		
9f2c...				
Owner	alice@acme (user)	Not after 2026-06-26 12:00Z		
(23 h)				
OS	macos	Status	valid	
✓				
Overlay	fd7a:helix:1::2	Auto-renews over control		
channel				
Enrolled	2026-06-20			
WG pubkey 32B	Yg8...== (public)	[Force re-		
issue]				
Presence / health (no traffic data –				
C3)				
Last seen	online (live)	Transport	masque-h3	RTT 23
ms				
Groups				
group:admins				
@ HelixVPN stores identity & policy only. There is no connection,				
destination, or bandwidth record for this device, by construction.				

Revoke flow (§11.4.66 confirm). [Revoke ⌵] opens a confirmation dialog ("Revoke alice-mbp? Its tunnel is force-closed and its peer is removed from every map within ~1 s [CP §9.3]. This cannot be undone — re-enrollment is required."). On confirm: POST /v1/devices/:id/revoke (admin-only [CP §8.1]); the row flips to revoked, an audit device.revoked row is written, and the topology graph drops the node live. Revoke is **admin-only**; operator/member see it disabled with a tooltip stating the required role.

States. *Loading* — header skeleton + panels shimmer. *Error* — per-panel inline error (cert panel can fail independently of identity). *Revoked device* — the whole header tints with the danger/revoked treatment and the revoke button becomes a disabled "Revoked" label; re-enroll guidance is shown.

A11y / light-dark. The connector "advertised prefixes" sub-panel (for connectors) links to IPAM. Panels are surface.raised; "valid ✓" is feedback.success, an expiring-soon cert (< 1 h) is feedback.warning. The privacy note is text.secondary, present on **every** device surface so the no-logging guarantee is never out of sight.

6. Users & SSO (OIDC config, roles)

Purpose. Manage the human roster and how they authenticate. Two halves: **Users & roles** (the users table — admin > operator > member [CP §2.2/§8.1]) and **SSO configuration** (the OIDC IdP the tenant trusts — Keycloak/Authentik [CP §9.1]). Admin-only.

Users & SSO				[+ Invite]	
user]					
Users					
NAME / EMAIL		IDENTITY	ROLE	DEVICES	
STATUS					
alice@acme		OIDC	admin ▼	3	
active					
bob@acme		OIDC	operator ▼	2	
active					
carol@ext		OIDC	member ▼	1	
active					
(device-token user)		anonymous	member	1	no
PII (C6)		← anon user row			
SSO (OIDC)					
Provider		Keycloak		Status ●	
connected					
Issuer URL		https://id.acme.example/realms/			
acme					

Client ID	helix-console	
Client secret	 [rotate]	(never displayed/logged)
Group claim groups	→ maps IdP groups to Helix groups	
[Test connection]		
[Save]		

Interactions. Inline **role editor** per user (a dropdown that writes the new role; an admin cannot demote the last remaining admin — guarded with an explanatory disable). **Invite** mints an OIDC-bound invitation or, in privacy mode, points to the anonymous device-token path. **SSO config** edits the issuer / client-id / claim mapping; the **client secret is masked, never rendered or logged** (§11.4.10) and only ever rotated. [Test connection] performs a live OIDC discovery probe and reports the honest result.

States. *Loading* — roster skeleton. *Empty* — a single bootstrap admin + "no SSO configured yet — using local/bootstrap auth" guidance. *Error* — a failed SSO test shows the precise discovery/JWKS error (feedback.error), never a vague failure. The **anonymous user row** is rendered with a "no PII" badge to make the privacy posture (C6) visible, not hidden.

A11y / light-dark. Role dropdowns are keyboard-operable; the secret field has an aria-label and a copy-suppressed value. Status dots ("connected") pair with text. Light: surface.default/ surface.raised; the rotate/secret controls use feedback.warning emphasis to signal sensitivity.

7. Policy editor (ACL rules, compile / preview, fail-closed)

Purpose. The Console's highest-stakes screen — author the declarative ACL document (policies.spec jsonb [CP §7.1, POL §2]), **dry-run compile** it, inspect the **effect-diff** vs the active version, and **activate** (with instant rollback). This is where "1 user → N networks" becomes compiled per-node visibility (default-deny, fail-closed [POL §0.1]).

Layout (three-pane: editor | effect-diff | version history).

Policy editor	active: v8	editing: v9 (draft)	⊗
not saved			

SPEC (declarative ACL)	EFFECT PREVIEW (dry-run)
VERSIONS	
draft ●	Compile: ✓ ok (0.04 s) v9
groups:	v8
active ✓	
group:admins: [...]	+ admins → office-lan:*
v7 ↺	
hosts:	+ admins → warehouse-cams
v6 ↺	
warehouse-cams:	- contractors → office-lan
v5 ↺	
10.10.0.0/24	(removed from prior draft)
...	
acls:	[diff
v8↔v9]	
- action: accept	Visible-peer changes:
src: [group:admins]	alice-mbp +2 -0
dst: ["*:*"]	carol-lap +0 -1
- action: accept	
src:[group:contr.]	△ advisory: warehouse-cams
dst:["warehouse-	overlaps office-lan CIDR
cams:554,80"]	→ resolved by 4via6
exitNodes:[group:adm]	
	[Validate] [Activate
v9]	
x ERROR (fail-closed): host "cam-net" dst CIDR not covered by any	
← blocking error state	
advertised prefix → fix in IPAM, or correct the host. [go	
to IPAM]	

Components. PolicyEditor [CLIENT §7.2] — a structured text/JSONC editor with syntax highlighting of the ACL grammar [POL §2.3]; the **effect-diff** pane shows the dry-run CompiledPolicy delta vs the active version (added/removed edges, per-device visible-peer +/- counts [POL §4.3]); a **version history** rail listing every policies.version with the active one marked and each prior one **one-click re-activatable** (instant rollback [POL §6, CP §7.4]).

Fail-closed behaviour (the load-bearing UX). Activation is **disabled** until a clean dry-run [POL §5]. Blocking errors (unknown group/host; a dst CIDR not covered by any advertised_prefixes; a

rule granting a *revoked* device; an `exitNodes` entry resolving to a connector) render in the bottom error bar with the **exact** offending token and an actionable link (e.g. "→ go to IPAM") — never a generic "invalid". Advisory findings (overlapping-CIDR `route.conflict`) appear as warnings in the preview and **do not block** activation [POL §5.1, REG §6.3].

Activation flow. [Activate v9] → confirm dialog summarising the effect-diff ("activate v9: 2 devices gain access, 1 loses access; convergence ~<1 s, no restart [CP §10.1]"). On confirm → `POST /v1/policies/{v}/activate`; the active marker moves to v9, an audit `policy.activate` row is written, and the topology + device maps reconcile live. A bad activation is rolled back by re-activating an older version from the rail (instant, no recompile [POL §6.3]).

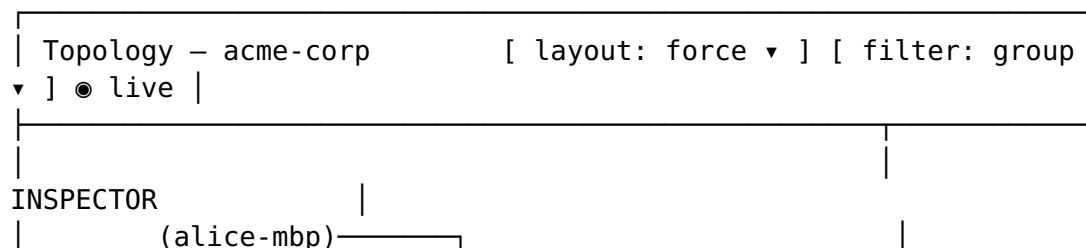
States. *Loading* — editor + panes skeleton; the active version loads first. *Empty* — a brand-new tenant shows a **starter template** (default-deny + a commented example) rather than a blank document. *Compiling* — the preview shows a "compiling..." state with the elapsed timer; a clean result flips to the diff. *Error* — blocking compile errors (red bar, activate disabled); the editor highlights the offending line. *Dirty* — an unsaved draft shows "∅ not saved" and warns on navigate-away.

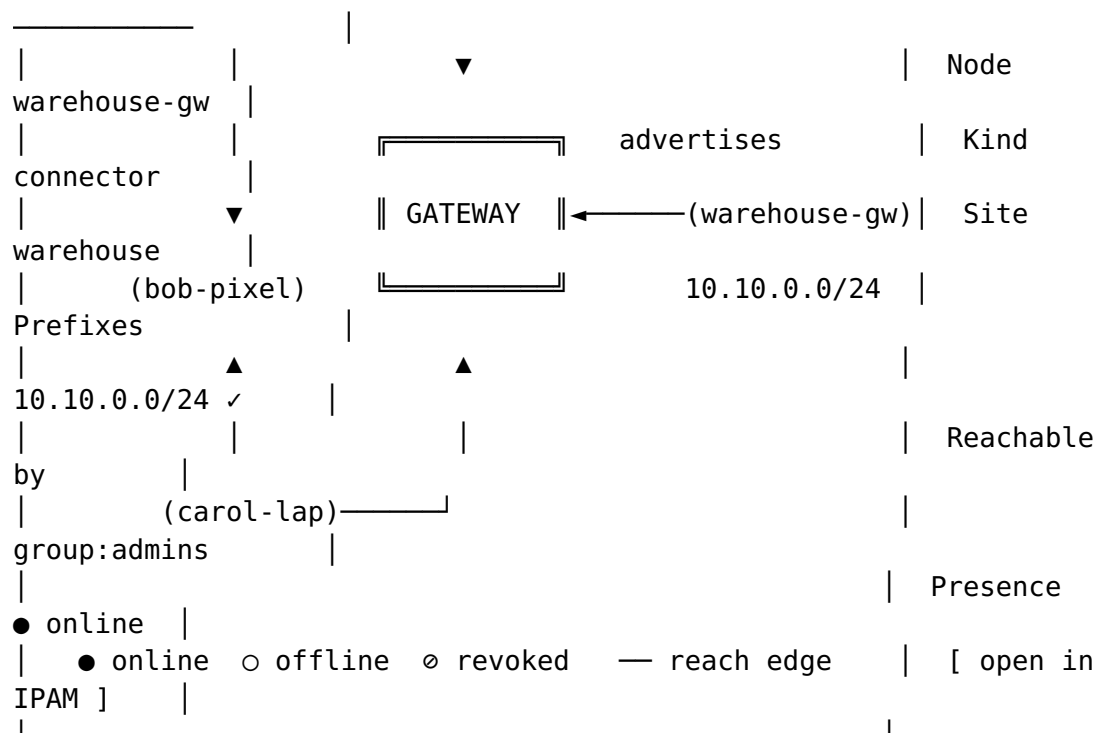
A11y / light-dark. The editor is fully keyboard-navigable with a screen-reader description of compile status (announced, not colour-only). Added edges are `feedback.success` green, removed are `feedback.error` red, advisory is `feedback.warning` amber — each with a `+/-/△` glyph so the diff is legible without colour (CI5). Light: editor on `surface.sunken` (code well), panes on `surface.raised`; dark mirrors with the same roles.

8. Topology (the coordinator graph)

Purpose. Visualise the tenant's overlay as the **coordinator's in-memory topology graph** [COORD §1] — devices, connectors, advertised prefixes, group membership, and *who-can-reach-whom* edges as compiled by the active policy (need-to-know, C4). It is the spatial companion to the tabular Devices/Policy views and updates live.

Layout (graph canvas + inspector).





Components. A graph canvas (UNVERIFIED: the concrete graph-render library is pinned by the component spec — stated here as a force/hierarchical layout choice, not a fixed lib). **Nodes** = devices/connectors/gateway, coloured by presence (connection-state palette, green/grey/red, **with a glyph + label**, never colour-only). **Edges** = reach relationships from the compiled policy (visible[d] [POL §4.1]). An **inspector** pane shows the selected node's identity, prefixes, who-can-reach-it, presence, and deep-links (connector → IPAM, device → device detail).

Interactions. Click a node → inspector + cross-links; pan/zoom; filter by group or kind; layout toggle. Live: device.online/offline/revoked, connector.prefixes.changed, and policy.compiled events animate the graph (node colour change, edge add/remove) without refresh [COORD §4.2].

States. *Loading* — graph skeleton (placeholder node ring) while the snapshot hydrates. *Empty* — a 1-node graph (just the gateway) with "enroll a connector or client to grow your network". *Error* — if the snapshot fails, an inline retry over the last-known graph (marked stale). *Large-graph* — beyond a node threshold the view clusters by group/site and offers search-to-focus (so the canvas stays legible — no overlap, §11.4.162; nodes are spaced and labels never overlay each other).

A11y / light-dark. A **table fallback** of the same graph data is available (the graph is reinforcement, not the only way to read topology — keyboard/SR users get the tabular Devices + a reach

matrix). Canvas background is background.canvas; edges use border.strong; the danger/revoked node uses the red state colour at z-top so it is never occluded [COLOR §3.2/§5].

9. Audit log (control-action audit — no traffic logs)

Purpose. The tamper-evident record of **control actions** — device enroll/revoke, policy create/activate, prefix changes, user/role changes, cert issue/revoke (audit_events, closed AuditAction vocabulary [TEL §4.2]). The screen's defining feature is what it **cannot** show: there is **no** connection, destination, flow, or bandwidth record — by construction (C3 [CP §0.1/§2.4]).

Audit [action ▼] [actor ▼]			
[2026-06-25 range]			
@ This is a CONTROL-action log. HelixVPN keeps NO traffic, connection, ← the no-traffic banner			
destination, or bandwidth records – that data does not exist (C3).			
TIME (UTC)	ACTOR	ACTION	TARGET /
META			
12:04:31	alice	device.revoke	dev-31 (old-laptop)
12:03:10	system	policy.activate	v8 → see Policy editor
12:01:55	bob	device.enroll	connector warehouse-gw
11:58:02	carol	prefixes.change	+10.20.0.0/24 on conn-B
11:40:18	alice	user.role.change	carol member→operator
...			
< newer older > export: CSV ▼ (control actions only)			

Components. A reverse-chronological feed (indexed audit_events (tenant_id, ts DESC) [CP §2.2]); **filters** by closed action vocabulary + actor + time range; deep-links from a row's target to the relevant screen (a policy.activate row → that version in S4; a device.revoke row → S2d). Export is **CSV of control actions only** — the export pipe shares the no-traffic guarantee.

The privacy banner is load-bearing. It is always visible at the top of S7 and restated near the export control: the Console is *incapable* of producing a traffic log because the schema holds none (the §2.4 schema-lint is the runtime proof [CP §2.4]). This turns "we don't log" from a promise into a visible product property.

States. *Loading* — feed skeleton. *Empty* — "no control actions in this range" (a legitimate empty, not an error). *Error* — banner + retry over last-known rows. *Read authz* — audit **read** is admin-gated [TEL §4.5]; operator/member see a "audit is admin-only" state, not a partial leak.

A11y / light-dark. The feed is a semantic table; rows are keyboard-navigable; the privacy banner uses feedback.info (blue) so it reads as informational, not alarming. Light/dark via standard surface/text roles.

10. IPAM / prefixes

Purpose. The address plane: the tenant's overlay **ULA /48** pool [IPAM §1, CP §3] and the **advertised prefixes** each connector exposes [REG §6], including the **overlapping-CIDR conflict** surface and the 4via6 site-disambiguation that resolves it (D4). Operator+.

IPAM / prefixes – acme-corp			
Overlay pool			
ULA /48	fd7a:helix:alb2::/48	allocated hosts	
144			
Gateway	fd7a:helix:alb2::1	next	
host	::92		
⚠ /48 capacity is astronomically large; hosts are never recycled (IPAM).			
Advertised prefixes (by connector)			
CONNECTOR	SITE-ID	CIDR	ENABLED
STATUS			
warehouse-gw	site-1	10.10.0.0/24	✓
ok			
office-gw	site-2	192.168.50.0/24	✓
ok			
branch-gw	site-3	192.168.50.0/24	✓
office-gw	← conflict row		
Conflicts			
⚠ 192.168.50.0/24 advertised by office-gw (site-2) AND branch-			

```
gw (site-3) |
|   Advisory – not blocking. Resolved by 4via6 site
disambiguation: |
|   office → fd7a:helix:a1b2:0002::/96   branch →
...:0003::/96 |
|   [ accept 4via6 mapping ]   [ choose authoritative
connector ] |
```

Components. An **overlay-pool card** (ULA prefix, gateway address, next host, allocated count [IPAM §1.2]); an **advertised-prefix table** grouped by connector with site-id, CIDR, enable toggle, and per-row status; a **conflicts panel** listing overlapping-CIDR route.conflict.detected items [REG §6.2] with the **4via6 resolution** spelled out (the per-site /96 mapping the control plane ships in the network map [IPAM §4.3]) and the two operator choices (accept 4via6 / designate authoritative).

Interactions. Toggle a prefix enabled; edit a connector's CIDRs (writes the same path as the connector-side advertise, [REG §6.1] — links to the Connector app for the source). Resolve a conflict via the offered choices (§11.4.66 — never auto-resolved silently). Cross-link: the Policy editor's "host CIDR not advertised" error lands here (S4 ↔ S6).

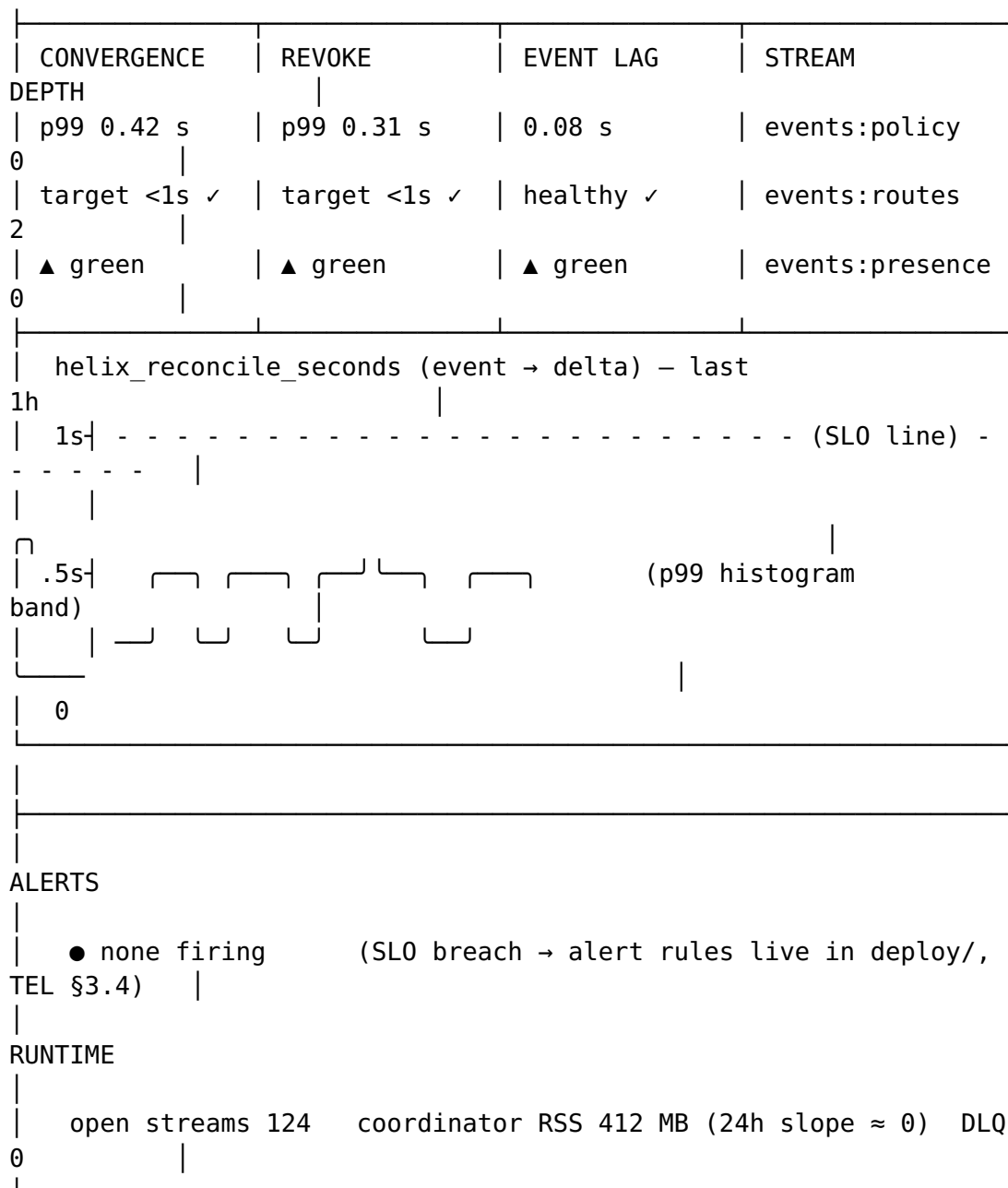
States. *Loading* — pool/table skeleton. *Empty* — "no connectors advertising prefixes yet". *Conflict present* — the conflict panel is highlighted (feedback.warning) and a count badges the nav item; it is **advisory** (never blocks policy activation [POL §5.1]). *Error* — per-section inline retry.

A11y / light-dark. Conflict rows carry a △ glyph + text (not colour-only); the 4via6 mapping is shown as monospace addresses (font.primitive.family.mono [DT §4]). Light/dark via surface/text/feedback roles; the conflict highlight is amber in both themes.

11. Telemetry / health (counters, convergence & event-lag SLOs)

Purpose. The operator's window into whether the control plane is **meeting its measured SLOs** [TEL §3, CP §10.2]: convergence p99 (event → delta-on-wire < 1 s), revoke latency (< 1 s), event lag, stream pending depth, and runtime health — all from Prometheus series the telemetry service emits [TEL §3.2/§3.3]. This is the anti-bluff "is it actually fast?" screen.

```
| Telemetry / health – acme-corp                               window:
1h ▼ |
```



Components. Four **SLO gauges** (convergence p99, revoke p99, event lag, stream depth) each green/amber/red against its target [TEL §3.4]; a **convergence histogram** with the SLO line drawn at 1 s so a breach is visually obvious; an **alerts** list (the *series* are emitted here, the *alert rules* live in deploy/ [TEL §3.4] — the screen states this honestly); a **runtime** strip (open stream count, coordinator RSS with the 24 h ≈0 slope soak signal [CP §10.2], DLQ depth `helix_events_dlq_total` [CP §5.4]).

Honest-SLO logic (§11.4.6). Every gauge shows the *measured* number and its target; "✓" appears only when measured < target. A breach turns the gauge red and surfaces in the alerts list — the screen never shows "healthy" without the figure that justifies it. This is the visible face of the anti-bluff < 1 s promise.

States. *Loading* — gauge/chart skeleton. *Empty* — a just-bootstrapped tenant with no traffic yet shows "no data in window — generate activity or widen the window". *Error* — if the metrics scrape fails, the screen says so (it does **not** paint green on absence of data — absence $\neq$ healthy). *Breach* — red gauge + alert row + a link to the offending series.

A11y / light-dark. Gauges pair colour with a value + ✓/△/✗ glyph; the SLO line is labelled. Charts have an accessible data-table fallback. Light/dark via standard roles; the SLO line uses feedback.info, breaches feedback.error.

12. Billing / subscription

Purpose. Manage the tenant's plan, seat/device usage against plan limits, and invoices. This is a **commercial** surface layered over the self-hostable core — its data source (the billing provider) is not specified by the control-plane docs.

Billing - acme-corp				
Usage		Plan		
Plan	Team	Devices		
142 / 250				
Renews	2026-07-20	Seats		
18 / 25				
Status	● active	Connectors		
7 / 10				
[Change plan]		[Manage payment method]		
57%				
Invoices				
DATE	AMOUNT	STATUS		
INVOICE				
2026-06-20	\$240.00	paid	[download	
PDF]				
2026-05-20	\$240.00	paid	[download	
PDF]				
...				

Components. A **plan card** (tier, renewal, status, change/payment actions); a **usage panel** (devices/seats/connectors against plan limits with a progress bar); an **invoice table** (date/amount/status/download). Approaching a limit shows a `feedback.warning` nudge; over-limit a `feedback.error` block on new enrollment.

D-CON-BILL-1 UNVERIFIED. The billing provider + its API are **not** specified by the control-plane docs [CP] — this screen's data source is a Phase-2 commercial decision (§11.4.66). The layout/states here are the *intended* shape; the provider binding is UNVERIFIED until that decision lands. The self-hostable MVP may ship the Console with Billing **hidden** behind a capability flag (no commercial backend) — also UNVERIFIED.

States. *Loading* — skeleton. *Empty / self-hosted* — when no billing backend is configured, the screen shows "self-hosted — no subscription" rather than a broken integration. *Error* — provider errors shown honestly with retry.

A11y / light-dark. Usage bars carry a numeric label; status dots pair with text. Standard surface/text/feedback roles, light + dark.

13. Settings

Purpose. Tenant-level configuration + per-admin preferences + the danger zone. Sectioned, with role-gated controls.

Settings	
► Tenant template	name, control-plane URL, default policy
► Appearance (admin)	theme ☒ system ☐ light ☐ dark (per-admin)
► Session TTL	signed-in devices, sign-out-everywhere, session
► Transport	default escalation ladder order (plain-udp→lwo→masque-h3)
► Notifications	which control events page the live rail / send alerts
► Danger zone	rotate tenant CA · delete tenant (admin-only, confirmed)

Components. Collapsible sections. **Appearance** is the per-admin theme toggle (system/light/dark — both themes first-class [CLIENT §7.1]). **Transport** edits the default `TransportPolicy.order`

ladder the control plane ships to agents [CP §4 TransportPolicy]. **Danger zone** holds irreversible, admin-only, double-confirmed actions (rotate the tenant CA key [CP §9.3] — the one true secret; delete tenant) each gated per §11.4.66 with a typed-confirmation dialog.

States. *Loading* — section skeletons. *Error* — per-section inline. *Role-gated* — operator/member see read-only or hidden danger-zone controls with the required-role tooltip.

A11y / light-dark. Every toggle is keyboard-operable and labelled; the theme toggle previews live. Danger-zone actions use feedback.error emphasis and require typing the tenant name to confirm. Standard roles, light + dark.

14. Cross-screen states, a11y & light/dark contract

These hold for **every** Console screen (so each section above states only its specifics):

Concern	Contract	Source
Loading	skeleton placeholders (shimmer), never a bare spinner on a blank screen; the most-important data loads first	original UX
Empty	a purposeful empty state with the next action (enroll, configure, invite) — never an ambiguous blank	original UX
Error	the honest cause (provider/JWKS/scrape error), an inline retry, and last-known data marked stale — never a generic "something went wrong"	§11.4.6
Stale/live	the live-stream chip goes amber on WS drop and lists show "may be stale — reconnecting"; the UI never paints fresh data it cannot prove is fresh	CI2 [CLIENT §0.1]
Confirm	every destructive control action (revoke, deactivate, rotate CA, delete) is double-confirmed via the interactive mechanism	§11.4.66
Colour ≠ only signal	every state dot/badge pairs colour with a glyph + text label	CI5 [CLIENT §0.1], [COLOR §0]

Concern	Contract	Source
No overlap / no label-overlay	spacing tokens keep $\geq$ space.scale.2 (8 px) between coloured regions; overlays use overlay.scrim, never a translucent brand wash; danger sits at z-top	§11.4.162, [COLOR §5]
Light + dark	every screen renders in both, all surface/text/feedback roles AA-proven both themes; the violet accent is the only per-app difference	§11.4.162, [COLOR §2/§4/§6]
Keyboard / SR	full keyboard nav (the Console is desktop-first); ⌘K command palette; data-heavy visuals (graph, charts) have table fallbacks	CI5, [CLIENT §7.3]
No traffic data	no Console screen can show connection/flow/bandwidth records; the Audit screen states this is by construction	C3 [CP §0.1/§2.4]

The violet light/dark accent (the Console's only brand difference).

Role	Light	Dark	Proof [COLOR §6]
action.primary / accent	violet.700 #6D28D9	violet.400 #A78BFA	white-on-violet.600 5.70 ; text 7.10 light / 6.47 dark — AA both

Everything else — surfaces, text, borders, **connection-state**, **feedback** — is the identical shared semantic set used by Client and Connector (D-COLOR-1); the Console never recolours a safety signal.

15. Surfaced decisions & cross-doc contracts

id	Decision / contract	Status
D-CON-1	Console is desktop-first (lg/xl home turf) but renders the full responsive ladder down to compact; the graph + policy editor + heavy tables	decided

id	Decision / contract	Status
	deferred-load so they never bloat a small layout.	
D-CON-2	Live data via GET /v1/stream WS/SSE folded into Riverpod; a visible stream-health chip + "may be stale" honesty when it drops.	decided [CLIENT §8.3]
D-CON-3	The Audit screen makes the no-traffic-log guarantee a <i>visible product property</i> (banner + control-actions-only export), backed by the §2.4 schema-lint runtime proof.	decided [CP §2.4]
D-CON-4	Policy activation is fail-closed: disabled until a clean dry-run; blocking errors name the exact token + an actionable link; advisory CIDR conflicts never block.	decided [POL §5]
C-CON-A (consumes)	All control-plane data + payloads owned by [CP §8] + the V3 svc-*.md docs; this doc owns only layout/state/interaction over them.	contract
C-CON-B (consumes)	Colour roles + violet accent owned by [COLOR]; token scales/breakpoints by [DT]; signature components by [component-library.md].	contract
C-CON-C (provides)	This screen set + per-screen states are the spec the visual-regression + golden suite asserts against (light+dark, every breakpoint, §11.4.162).	contract
U-CON-1 UNVERIFIED	Billing provider + API (S12) — a Phase-2 commercial decision (§11.4.66); the MVP may ship Billing hidden behind a capability flag.	open
U-CON-2 UNVERIFIED	The concrete topology graph-render library (S8) is pinned by the component-library.md spec; layout choice (force/hierarchical) stated here, library UNVERIFIED.	open
U-CON-3 UNVERIFIED	The exact GET /v1/stream event field set the live rail/dashboard digest renders is owned by [CP §8]; the digest's column choice is stated here, the wire shape UNVERIFIED against a frozen WS contract test.	open

Sources verified

- **Console flavor (admin, no core_ffi, Web-only build), capability model, AdaptiveScaffold, responsive law, %K palette, live-WS folding, theme-per-role, signature components (PolicyEditor/NetworkTile/...), a11y (colour ≠ only signal)** — final/03-client-core-and-ui.md §6 (flavors/capabilities), §7 (design system + responsive + a11y), §8.3 (Console live data), §0.1 (CI2/CI3/CI5) (read 2026-06-25).
- **Control-plane data every screen surfaces** — final/02-control-plane.md: §2.2 (devices/users/policies/audit DDL), §3 (IPAM ULA /48 + 4via6, D4), §4 (Coordinator proto + TransportPolicy), §7 (policy model/compiler, fail-closed, activate/rollback), §8 (REST + WS/SSE + RBAC), §9 (identity/enrollment/PKI/revoke < 1 s), §10.2 (measured SLOs), §0.1 C3 (no-logging by construction) + §2.4 (schema-lint) (read 2026-06-25).
- **Per-service screen data** — final/v03-control-plane/: svc-registry.md §2/§5/§6 (device model, presence, prefix conflict detection); svc-policy.md §2/§4/§5/§6 (ACL spec, compile, validation, activation/rollback, effect data); svc-coordinator.md §1/§4 (topology graph + event reactions); svc-telemetry.md §3/§4/§5 (Prometheus series, audit sink + closed AuditAction vocab, presence); svc-ipam.md §1/§4 (ULA address plan, 4via6 derivation); svc-identity.md (OIDC/anonymous identity modes) (read 2026-06-25).
- **Colour roles, the Console violet accent + its AA contrast proofs, the no-overlap / no-label-overlay colour rule, light+dark mandate** — final/v10-design/color-system.md §2 (semantic roles), §3 (connection-state palette), §4 (contrast proofs), §5 (no-overlap rule), §6 (per-app accent: Console = violet) (sibling, this wave).
- **Token tiers, spacing/radius/elevation/motion/z/ breakpoint scales (the responsive breakpoints + the surfaces/shadows the layouts use)** — final/v10-design/design-tokens.md §6 (scales), §1 (tiers) (sibling, this wave).
- **OpenDesign engine that emits the tokens** — final/v10-design/opendesign-foundation.md (sibling, this wave; §11.4.162).
- **Signature-component contracts (AdaptiveScaffold, PolicyEditor, NetworkTile, graph canvas) + the Connector/Client counterpart screens** — final/v10-design/component-library.md, .../screens-connector.md, .../screens-client.md (siblings; component-library + client-screens land in this same Volume-10 wave — cross-referenced, not duplicated).
- Items marked UNVERIFIED (U-CON-1 billing provider, U-CON-2 graph-render library, U-CON-3 frozen WS field set) are pending their named decision/contract test per §11.4.6 — **not** asserted as fact.
- **Layout, per-screen state machines, interaction flows, wireframes — NO external source needed — original HelixVPN UX design work** (the screen compositions are

owned by this document; they render the cited control-plane data through the cited design-system tokens/components).